CertiQ-Chirp — Diagram Atlas

Twenty-one reproducible diagrams covering architecture, data controls, model selection, quantum circuits, optimization, noise, business impact, and submission structure. Each asset is also supplied as PNG and SVG.

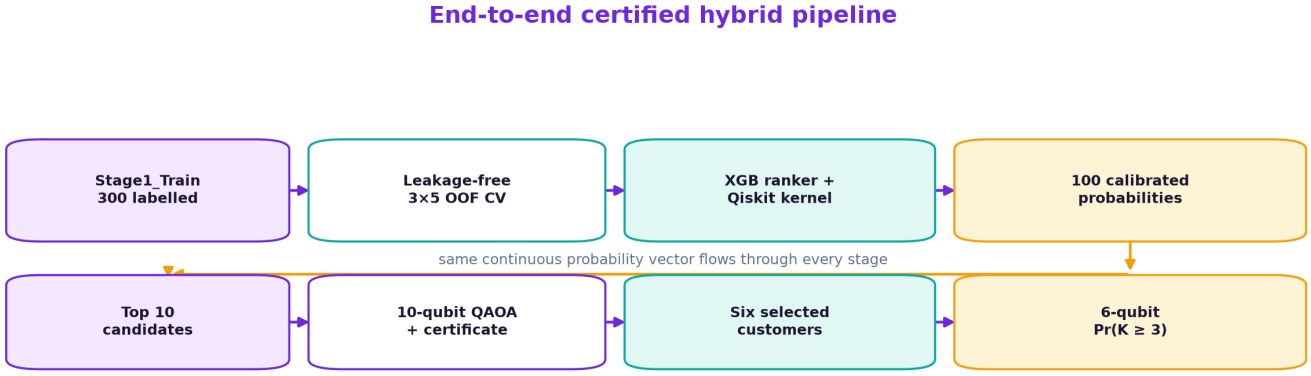
Figure 1. Cover artwork



Abstract cloud-quantum network and circuit motif used on the submission cover.

Section: Cover · Files: 00_cover_art.png, 00_cover_art.svg

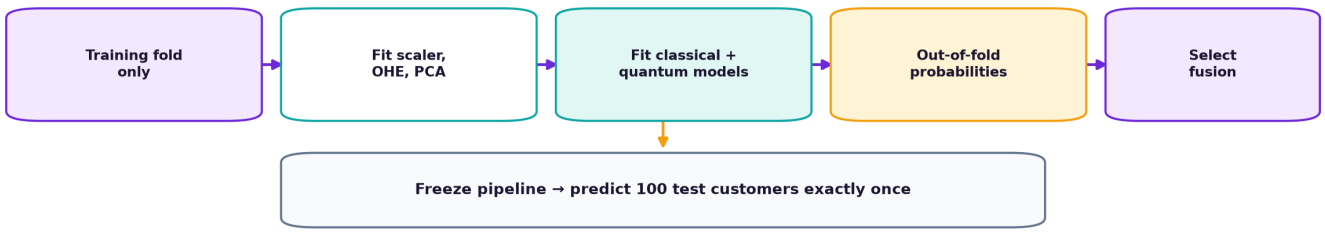
Figure 2. End-to-end architecture



Data, prediction, constrained selection, and quantum simulation form one traceable pipeline.

Section: Architecture · Files: 01_architecture.png, 01_architecture.svg

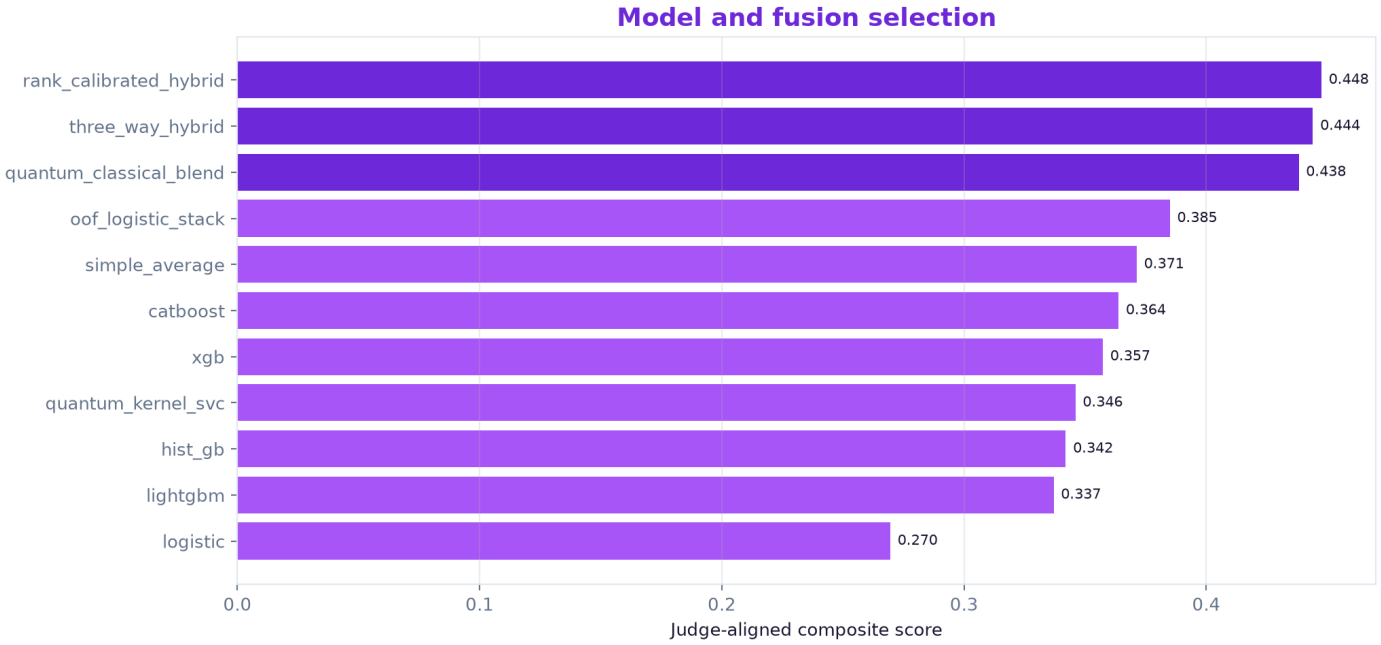
Figure 3. Stage 1 leakage controls



All preprocessing, calibration, and fusion weights are learned from out-of-fold training predictions.

Section: Stage 1 · Files: 02_stage1_pipeline.png, 02_stage1_pipeline.svg

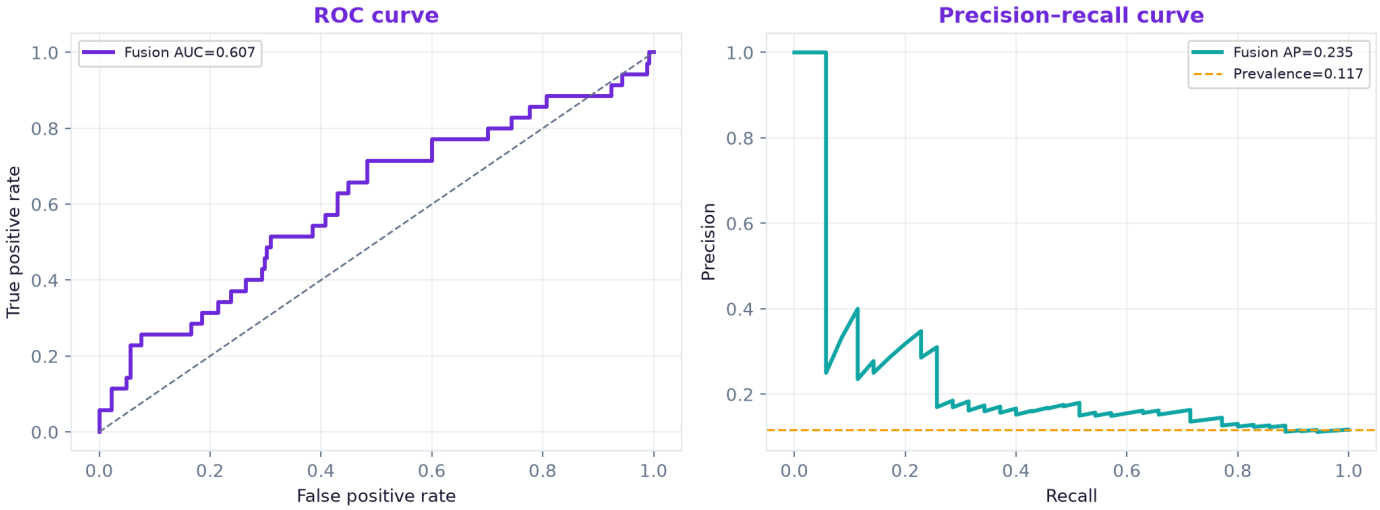
Figure 4. Model selection scoreboard



The 40% XGBoost / 60% quantum-kernel fusion achieved the highest judge-aligned composite.

Section: Stage 1 · Files: 03_model_scores.png, 03_model_scores.svg

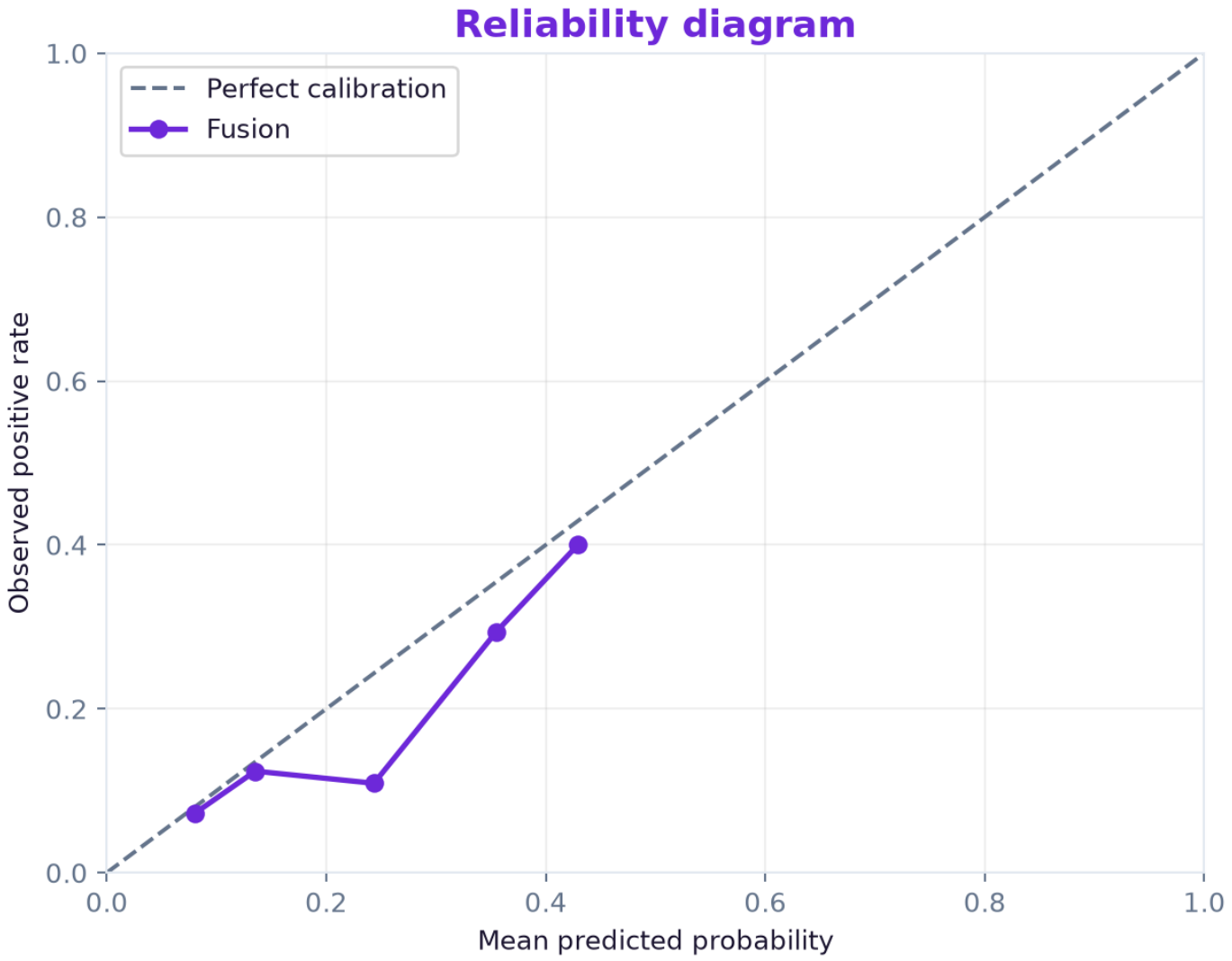
Figure 5. Discrimination curves



Out-of-fold ROC and precision–recall behaviour on the 300 labelled training customers.

Section: Stage 1 · Files: 04_roc_pr.png, 04_roc_pr.svg

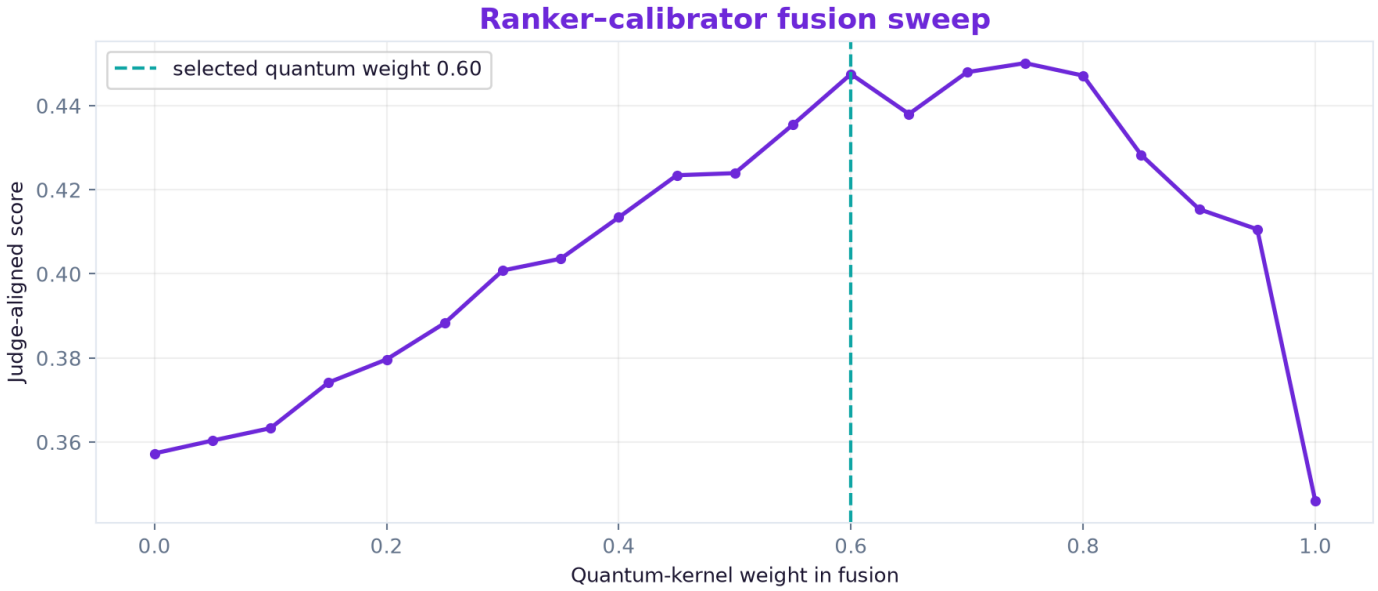
Figure 6. Probability calibration



Reliability curve supports the low Brier score, log loss, and expected calibration error.

Section: Stage 1 · Files: 05_calibration.png, 05_calibration.svg

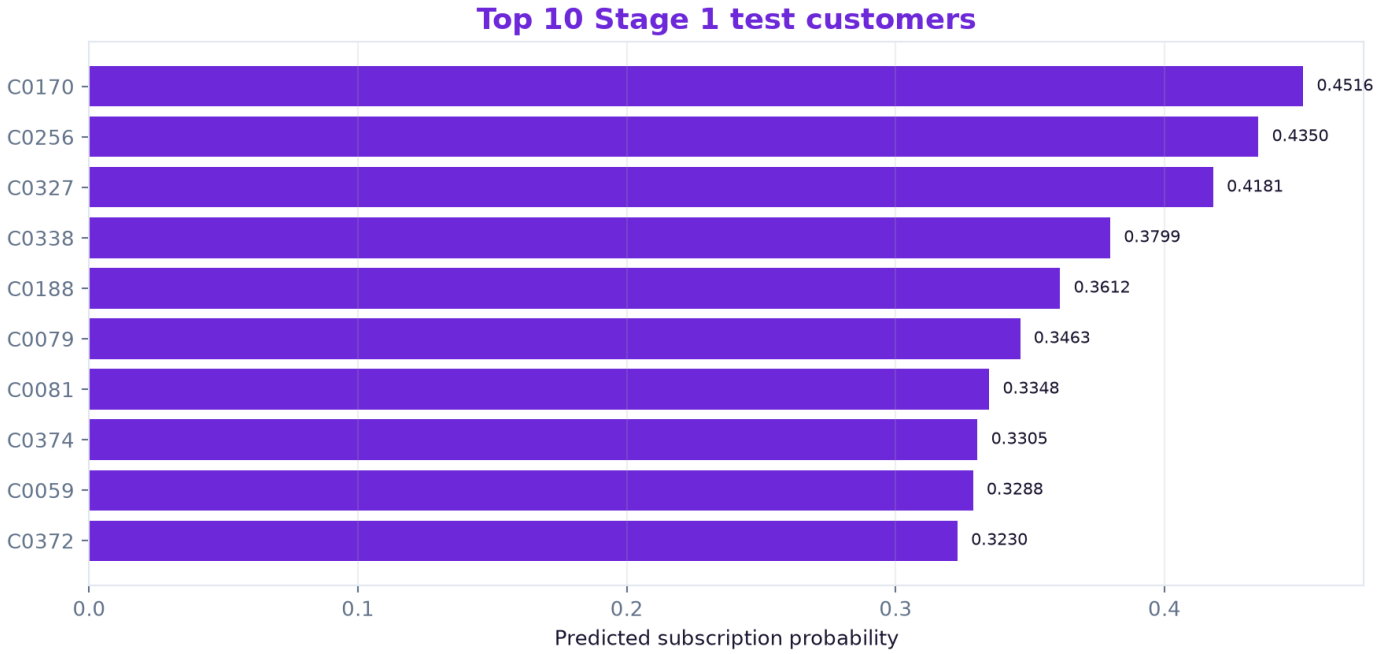
Figure 7. Fusion-weight sensitivity



The selected 60% quantum / 40% XGBoost point balances calibration with top-decile ranking.

Section: Stage 1 · Files: 06_fusion_sweep.png, 06_fusion_sweep.svg

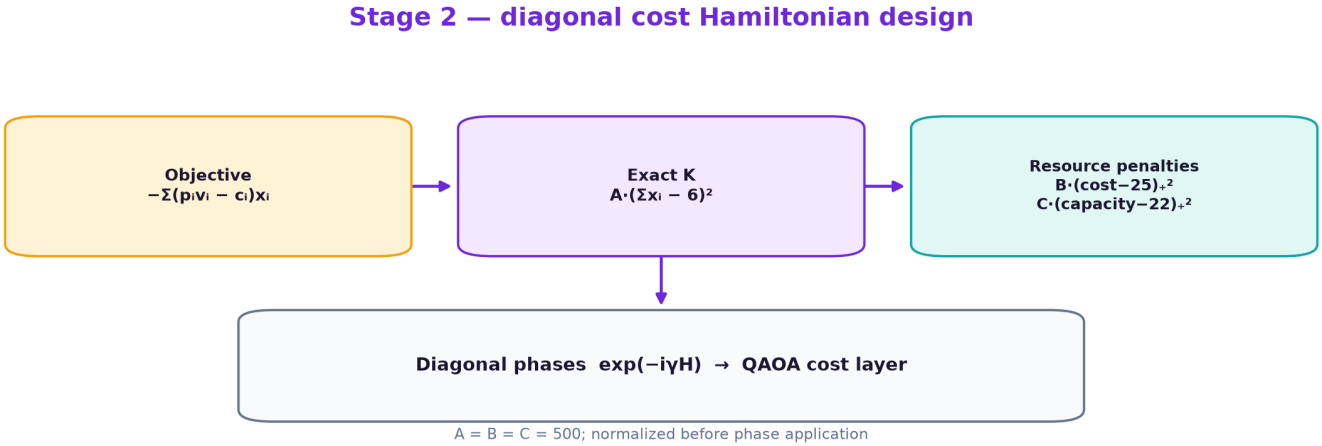
Figure 8. Top-10 candidate probabilities



Continuous probabilities are preserved and ranked with customer_id tie-breaking.

Section: Stage 1 · Files: 07_top10.png, 07_top10.svg

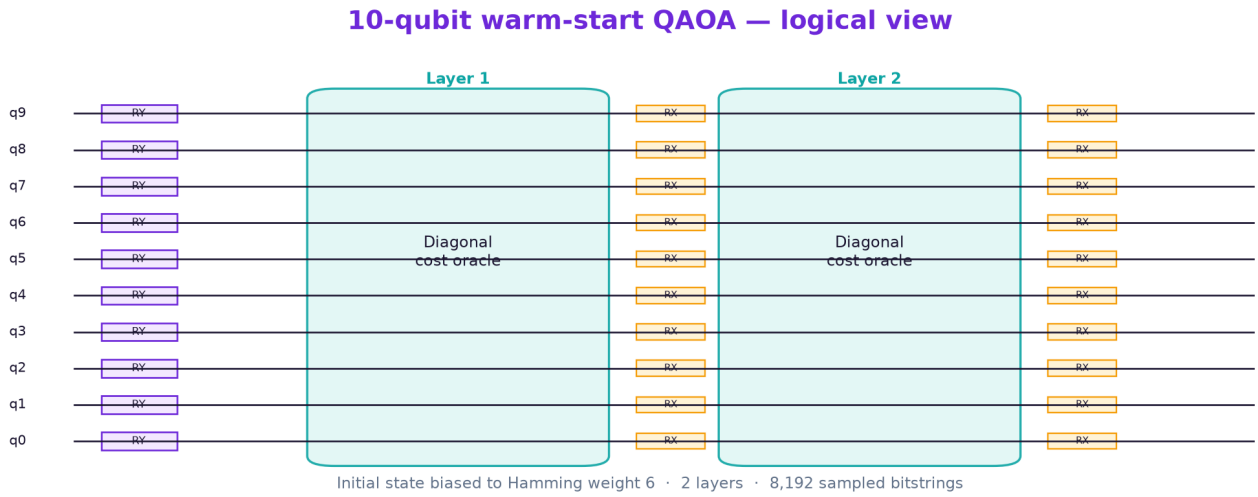
Figure 9. Constraint and penalty design



The exact-six and one-sided resource constraints are encoded in a diagonal QAOA cost oracle.

Section: Stage 2 · Files: 08_qubo_penalty.png, 08_qubo_penalty.svg

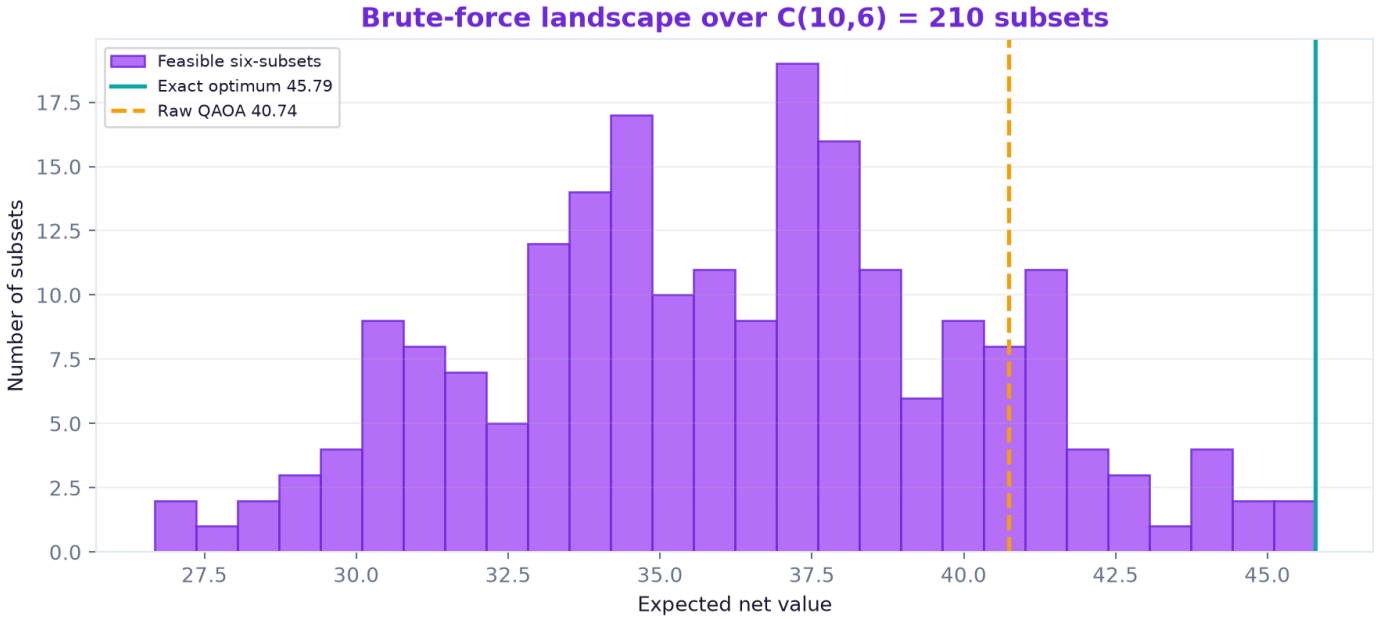
Figure 10. QAOA logical circuit



Warm-start RY initialization, two diagonal cost layers, and RX mixers on ten selection qubits.

Section: Stage 2 · Files: 09_qaoa_circuit.png, 09_qaoa_circuit.svg

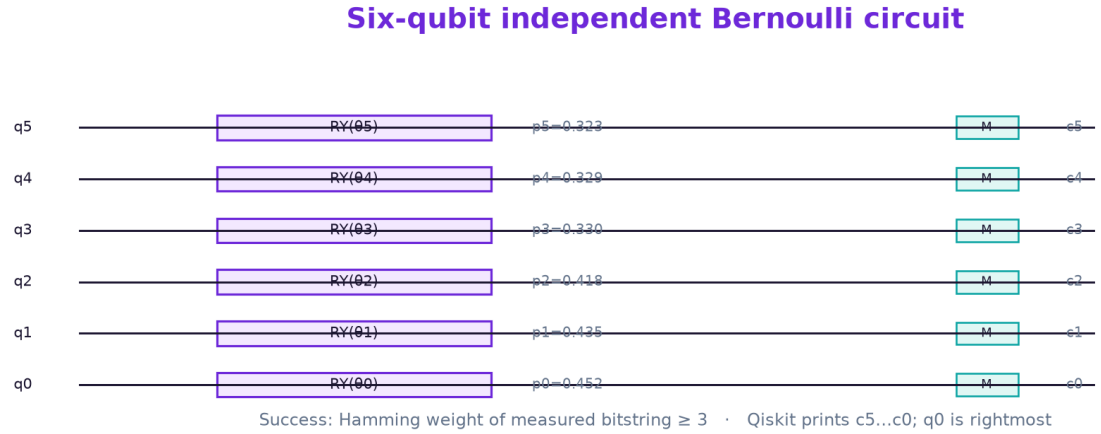
Figure 11. Stage 2 objective landscape



All feasible six-customer subsets are enumerated; raw QAOA and certified optimum are shown.

Section: Stage 2 · Files: 10_stage2_landscape.png, 10_stage2_landscape.svg

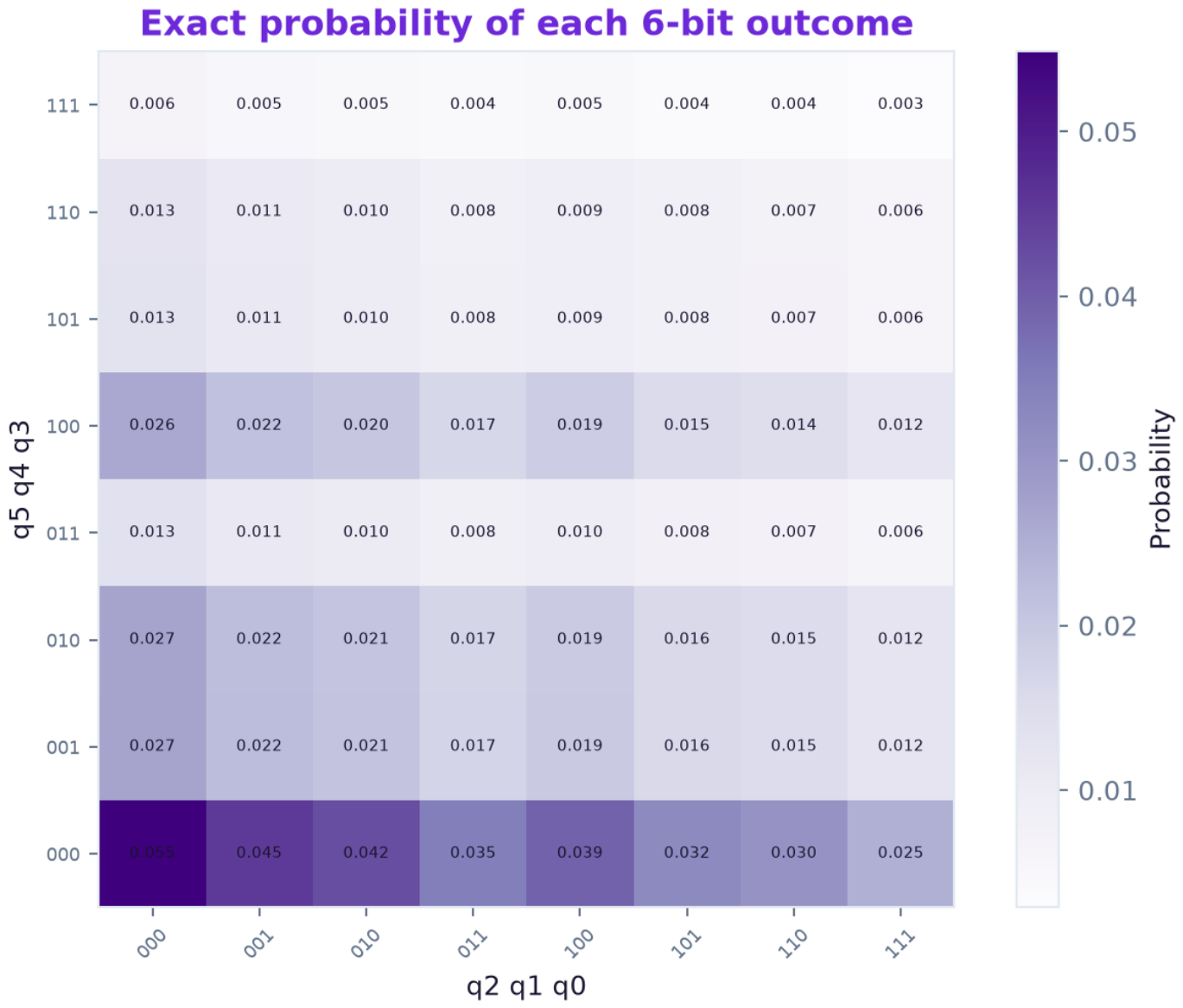
Figure 12. Stage 3 product-state circuit



One RY rotation per customer prepares the exact subscription probability on each qubit.

Section: Stage 3 · Files: 11_stage3_circuit.png, 11_stage3_circuit.svg

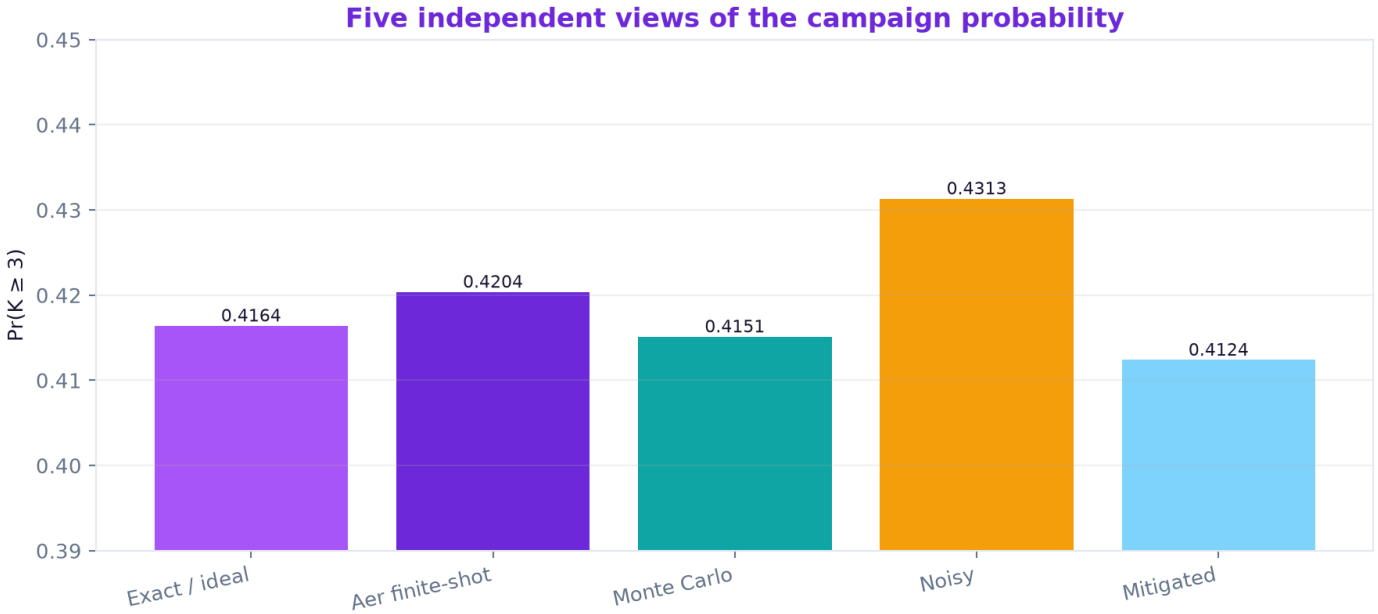
Figure 13. 64-outcome probability map



Independent Bernoulli outcomes; the upper-right region corresponds to three or more successes.

Section: Stage 3 · Files: 12_outcome_heatmap.png, 12_outcome_heatmap.svg

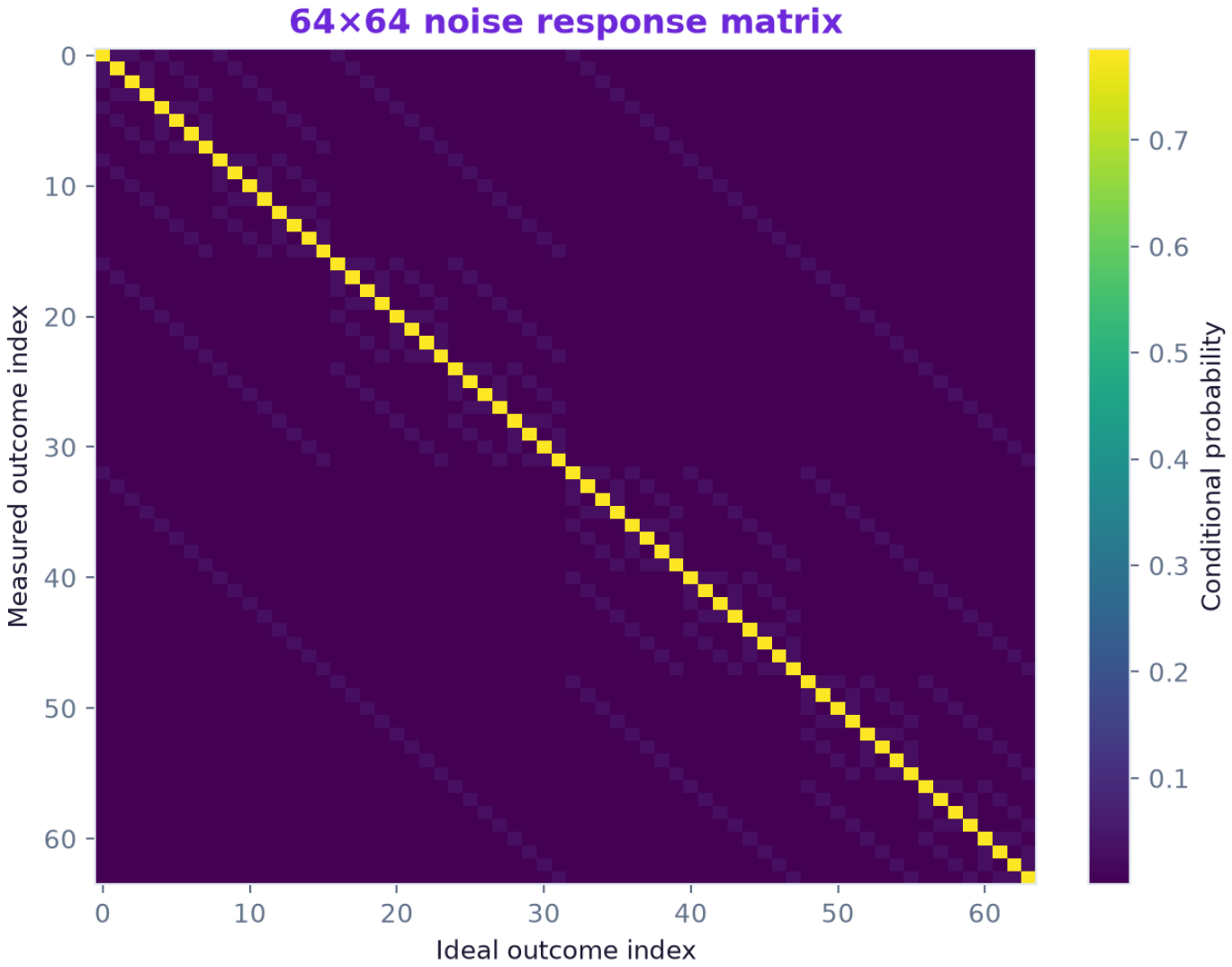
Figure 14. Stage 3 estimate comparison



Exact, ideal statevector, finite-shot, Monte Carlo, noisy, and mitigated results agree within reported errors.

Section: Stage 3 · Files: 13_stage3_estimates.png, 13_stage3_estimates.svg

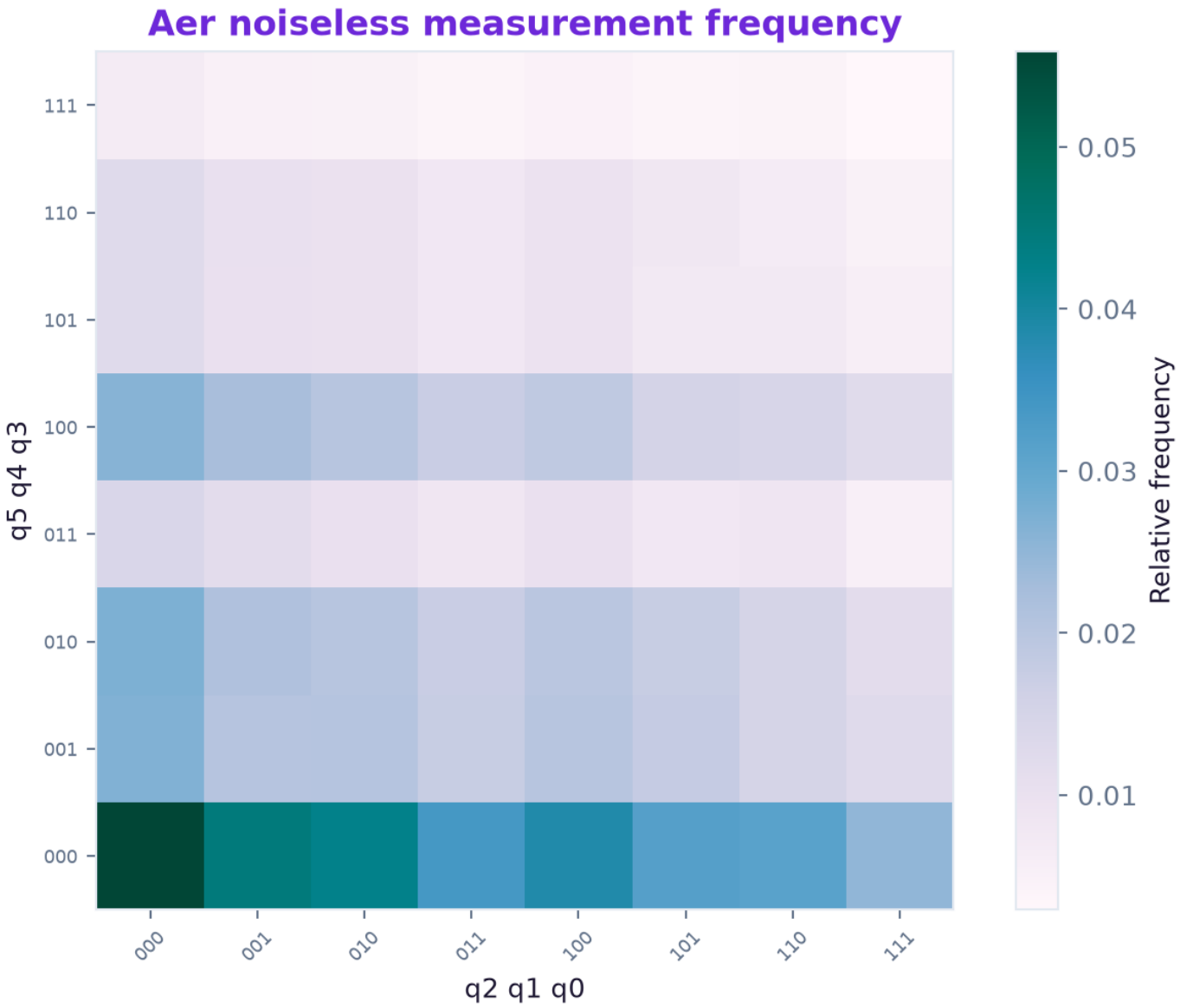
Figure 15. Noise response matrix



Tensor-product response of single-qubit depolarizing noise and symmetric readout error.

Section: Stage 3 · Files: 14_noise_response.png, 14_noise_response.svg

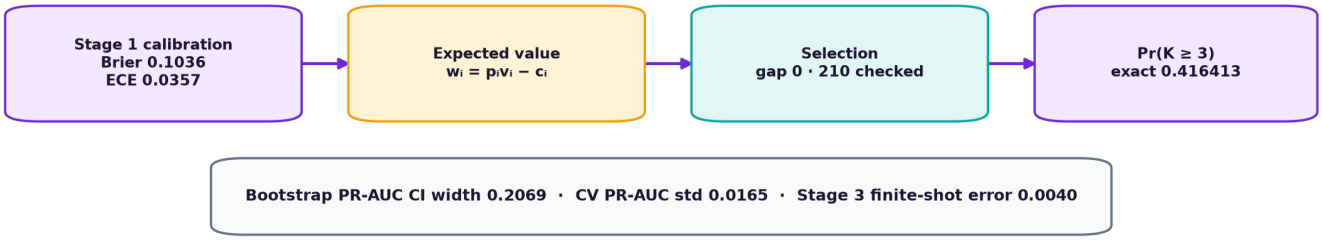
Figure 16. Measurement-count distribution



The 32,768 noiseless shots populate all 64 outcomes according to the product-state distribution.

Section: Stage 3 · Files: 15_counts_heatmap.png, 15_counts_heatmap.svg

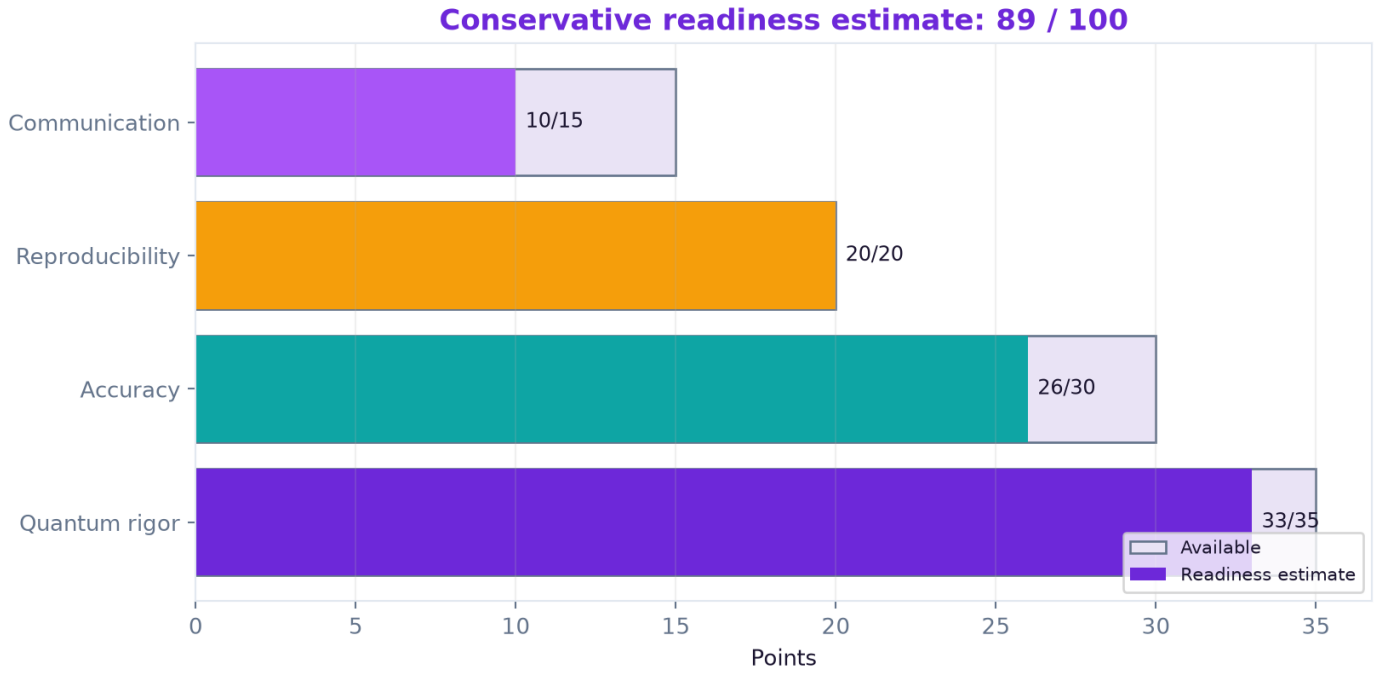
Figure 17. Error propagation



Probability uncertainty is carried into optimization and the final conversion estimate.

Section: Integration · Files: 16_error_propagation.png, 16_error_propagation.svg

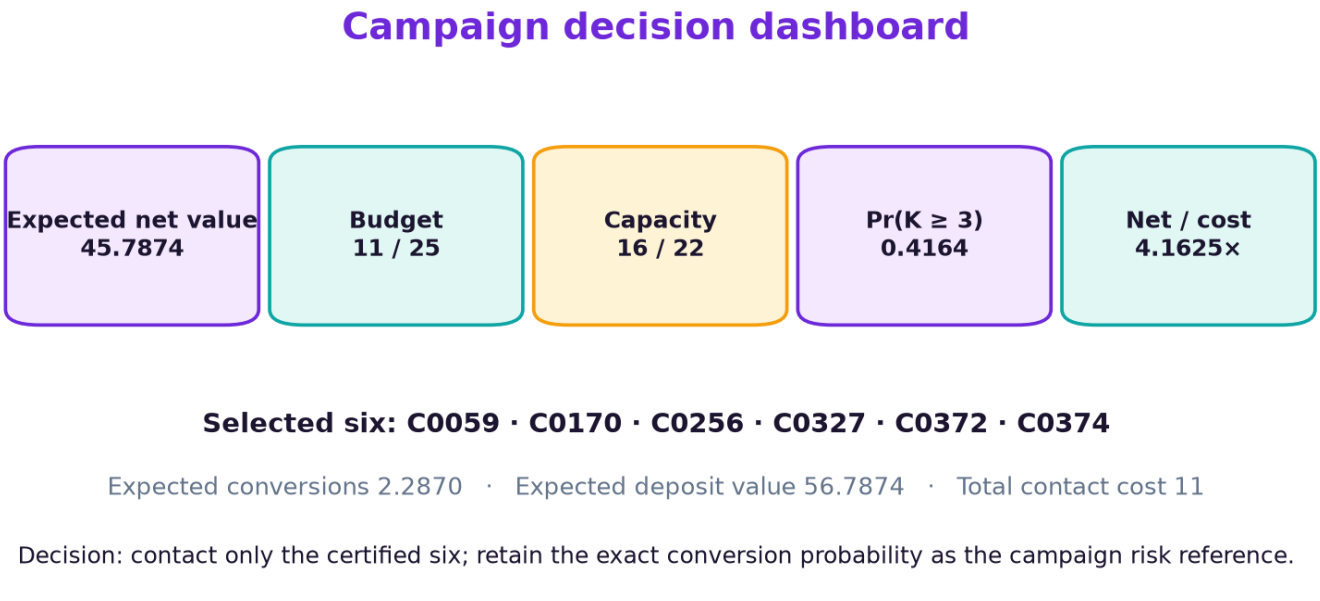
Figure 18. Judge rubric readiness



Rubric-aligned estimate based on quantum depth, exact references, reproducibility, and communication.

Section: Judging · Files: 17_rubric.png, 17_rubric.svg

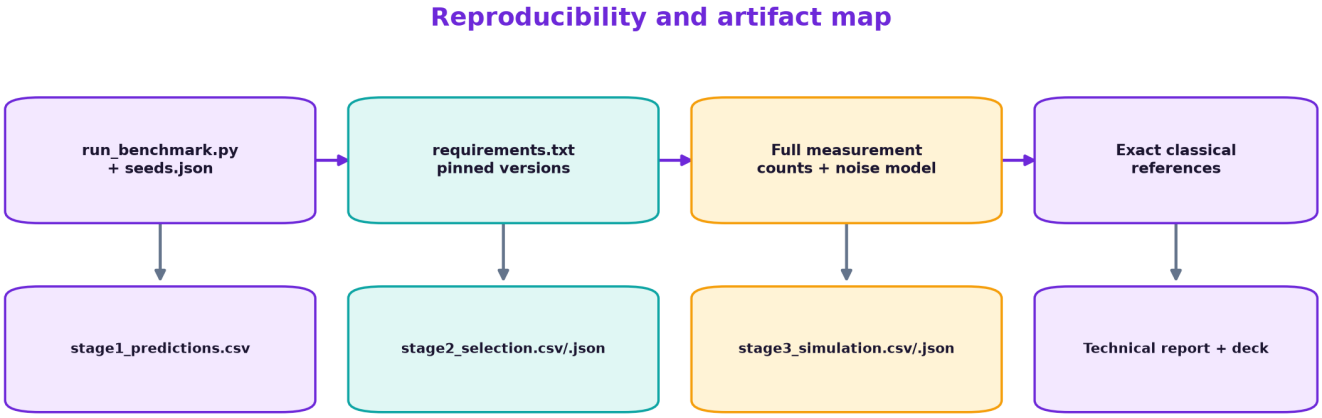
Figure 19. Business impact dashboard



Operational and financial interpretation of the final Stage 2 and Stage 3 results.

Section: Business · Files: 18_business_dashboard.png, 18_business_dashboard.svg

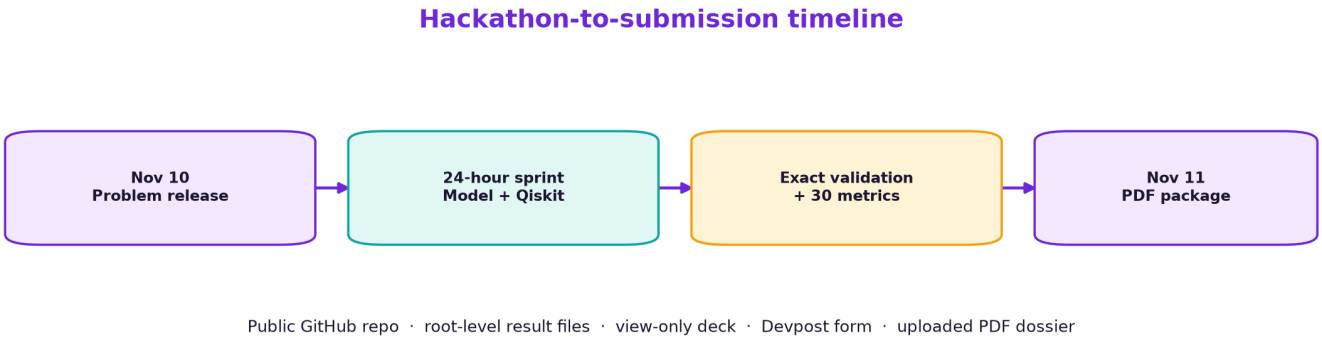
Figure 20. Reproducibility map



One runnable pipeline connects pinned inputs to every root-level deliverable and reference result.

Section: Reproducibility · Files: 19_reproducibility.png, 19_reproducibility.svg

Figure 21. Submission timeline



The final package consolidates code, result files, diagrams, and narrative into a judge-ready submission.

Section: Submission · Files: 20_timeline.png, 20_timeline.svg